Weekly Activity Report — AI Answer Safety Gate

Week ending: Friday 16 Oct 2026 (report prepared 14 Oct) **Team:** Search & Answering Infrastructure **In one sentence:** we tested four ways of checking whether an AI's retrieved documents actually contain the answer before it responds, and we now have a clear recommendation for which one to deploy.

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1. What we're building, in plain English

You already know how RAG works: when a user asks a question, the system *retrieves* a handful of relevant documents and hands them to a large language model, which writes an answer from them. That works well — **until the retrieved documents don't actually contain the answer.** The model then either makes something up or says "I don't know," and we have no way of knowing in advance which will happen.

Our project adds a **safety gate in front of the answer**. Think of it as an airport security checkpoint for every question:

- **Green light** → the documents clearly contain the answer; answer immediately and confidently.
- **Red light** → the documents are insufficient; refuse or escalate to a human.

The gate works by asking a small, very fast AI model one question per request: "*What is the probability these documents are sufficient to answer this question?*" — and only letting answers through when that probability is high enough. "Small and fast" matters: this checkpoint runs on *every* query, so if it's slow or expensive, the whole product slows down with it.

The single number that governs everything is our risk budget: we agreed the gate may let through a confidently-wrong answer on **no more than 5% of the questions it approves**. That 5% is the "false-sufficient rate" (FPR) you'll see throughout.

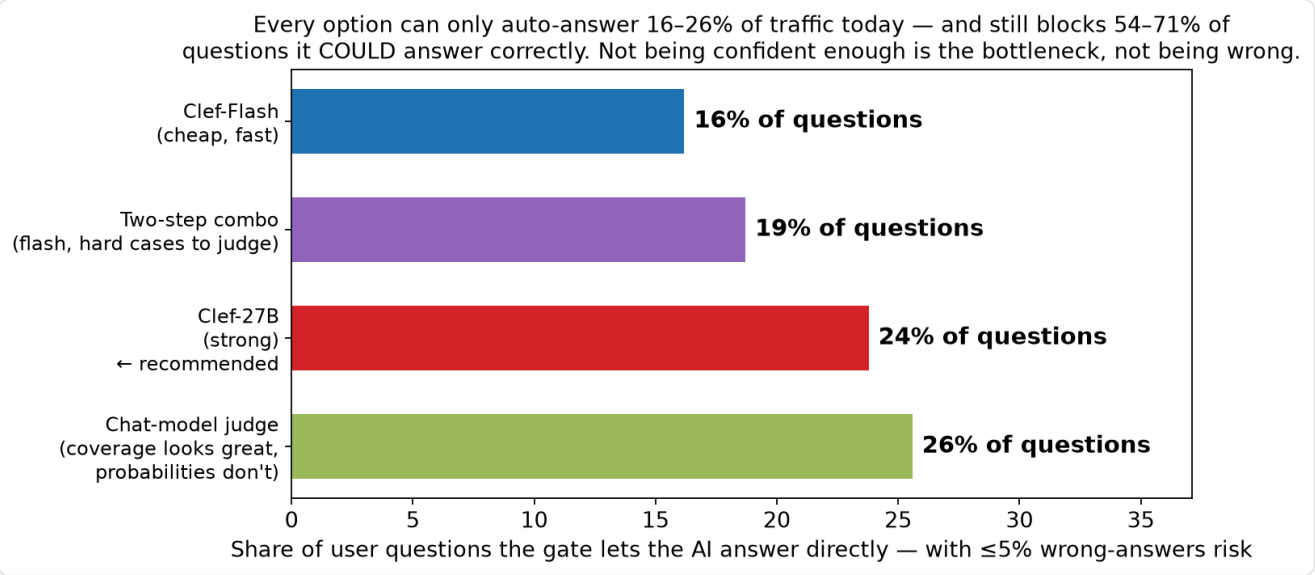
2. What we did this week

1. **Finished all benchmark runs — including the complete exam.** Every candidate model was tested on the full public SufficiencyBench exam (1,390 questions, each paired with a deliberately complete and an incomplete document set, so we always know the right verdict in advance). The final full run took 78 minutes and produced **13,900 judgements with zero failures**. All automated checks pass.
2. **Fixed a serious flaw in our two-step design** (see §4, "the combo"), re-ran it, and confirmed the fix works end-to-end — and still holds at full scale.
3. **Made the deployment decision** and stress-tested it on the complete dataset: this is the week we can say, with statistics behind it, which gate to ship.

3. The five contestants

Contestant	What it is	Analogy
Clef-Flash	Small (3B) purpose-built "instant judgement" model	Sharp junior analyst — fast, usually right
Clef-27B	Same idea, 9× bigger	Senior analyst — slower, noticeably more reliable
Chat judge	General chat model asked to referee each case	Consultant — talks a good game, overconfident
Two-step combo	Junior triages everything; only the hard 22% goes to the consultant	Junior + escalation
Baselines	Simple word-overlap and size checks	Checklists, no AI at all

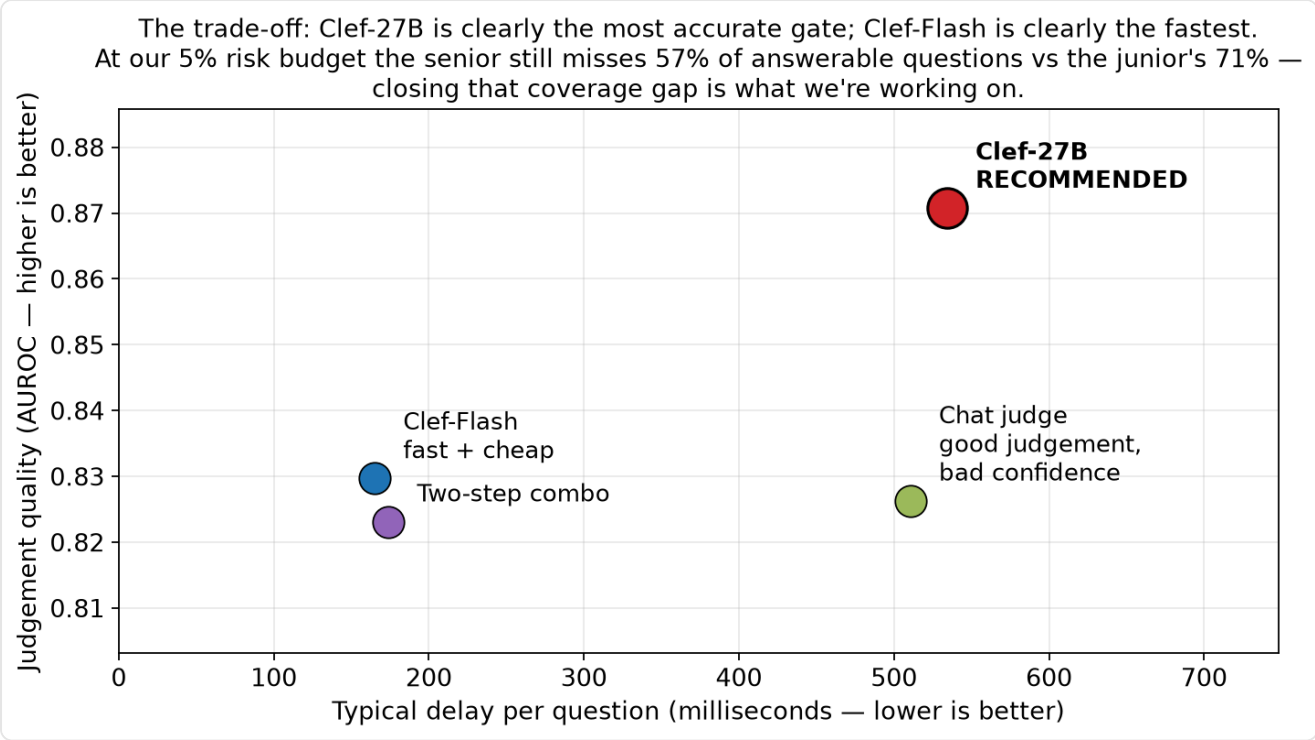
4. Results — the chart that matters



How to read this chart: each bar is one contestant; the length is the share of all user questions the gate would let the AI answer directly while keeping its error rate within our 5% budget. Longer is better — that's traffic we can serve automatically.

What it says (full dataset, 1,390 questions — these are final numbers, not a preview): - All four AI options can safely auto-answer only **16–26% of traffic** today. The rest gets blocked — and **54–72% of the blocked questions were actually answerable correctly**. The problem isn't that the gate lets wrong answers through; it's that it's too timid. - **The chat judge's bar looks the longest, and that's a trap.** It reaches 26% only by sounding artificially confident — when we audit *how sure it actually is* (see §5), it is the least trustworthy of all. On the full exam it now **breaks our 5% safety budget outright** (5.5%) — its extra coverage officially cannot be bought at the risk we accept. - **The two-step combo** (19%) does not beat simply using the senior analyst (24%) — the complicated pipeline isn't earning its complexity this week.

5. How each candidate performed — the scoreboard



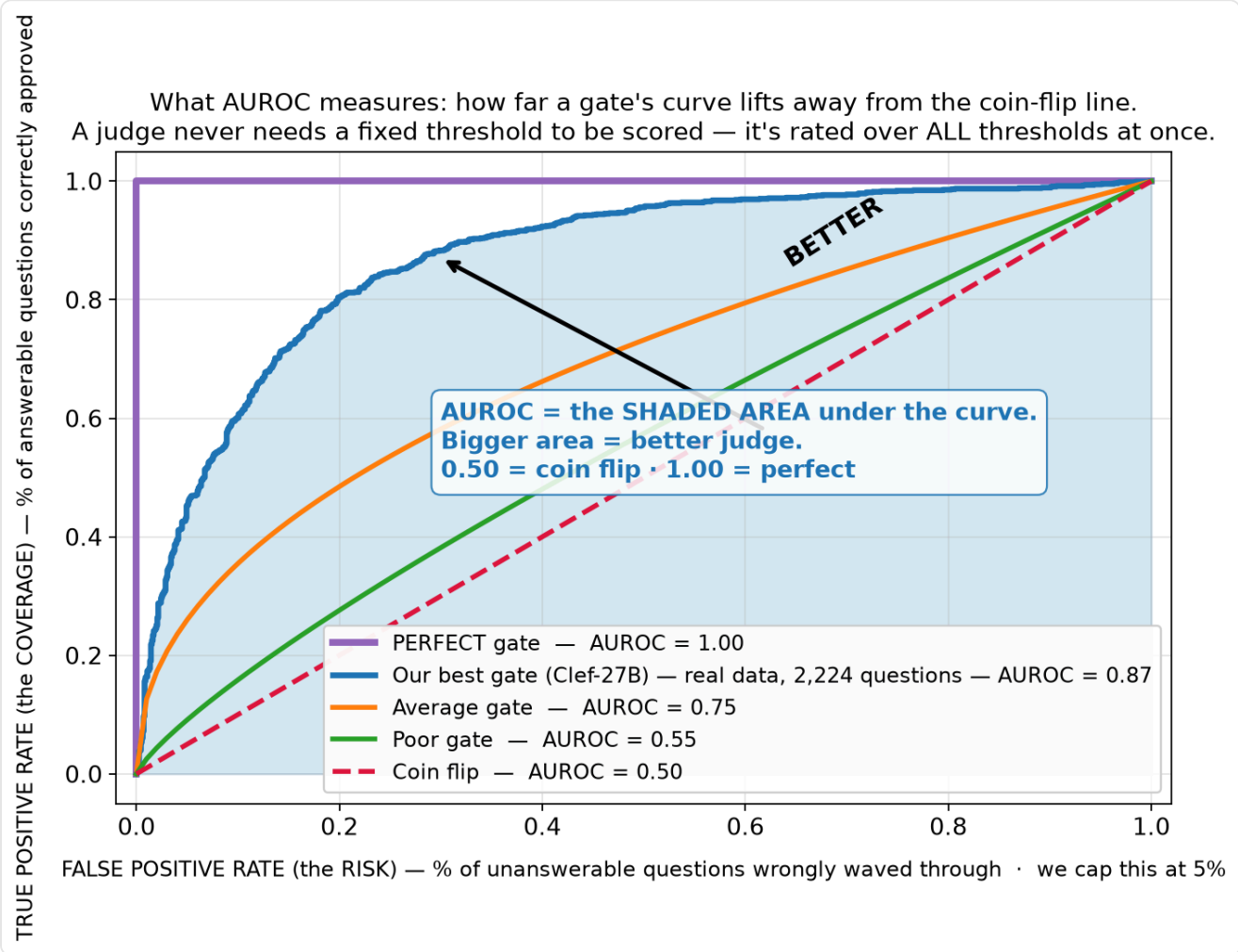
How to read this chart: every dot is one contestant. Right = higher judgement quality (AUROC, explained below); further up the axis = lower delay per question. There is no free lunch — you can usually have quality or speed, and the interesting question is who gets the best trade-off.

What it says: - **Clef-27B is the standout:** the most accurate judge by a clear margin, at a delay (half a second) that is acceptable for a pre-answer checkpoint. - **Clef-Flash is the speed king** (0.17 seconds) with decent accuracy — the budget option. - **The chat judge** sits near flash on quality but its confidence numbers are unusable (§6).

The scoreboard numbers, demystified

Term	Plain-English meaning
AUROC (0–1)	*Discrimination.* Give the judge a known-good case and a known-bad case at random — AUROC is the chance it rates the good one higher. 0.50 = coin flip; 0.88 = strong; 1.00 = perfect.
FPR (our 5% budget)	*The risk.* Of the questions the gate **approved**, how many actually had insufficient evidence — i.e. confidently wrong answers shipped to users. We cap this at 5%.
Coverage	*Throughput.* % of all traffic the gate lets through at that risk level. This is the business upside: coverage × traffic = auto-answered queries.
ECE (0–1, lower better)	*Calibration — "does its confidence match its accuracy?"* If the gate says "90% sure" on a batch of cases, it should be right ~90% of the time. High ECE means its stated confidence can't be trusted even if its gut decisions are partly right.
p50 / p95 latency	*Speed.* Typical response time (p50) and worst-case-for-95%-of-traffic (p95). We design around p95 because unhappy users remember the slow ones.
Significance (95% CI)	*Did it really win?* We re-scored the test thousands of times with reshuffled samples; if the measured gap holds in 95% of those re-runs, it's real, not luck. Clef-27B's lead is significant; the middle three are statistically tied.

What is AUROC? The 30-second version



That word — **AUROC** — is the one most likely to trip people up, so here it is with no jargon. *(The blue curve is the real, measured performance of our best gate on this week's full 2,224-question test set; the purple, orange, green and red curves are reference lines to calibrate your eye.)*

Every safety gate faces the same tension: be **strict** (few wrong answers slip through, but lots of good questions get blocked) or **relaxed** (lots of questions answered, but more wrong ones slip through). You can dial this strictness knob anywhere along a scale — a "threshold". Pick the strictest setting and you answer almost nothing; pick the loosest and you answer everything, correctly or not.

This chart shows *every possible setting of the knob at once*:

- **Right axis (the risk):** the share of unanswerable questions wrongly waved through.
- **Up axis (the payoff):** the share of answerable questions correctly approved.
- **Each point on a curve** is one strictness setting. The curve traces what happens as you slide the knob from ultra-strict (far left) to ultra-relaxed (far right).

The three reference lines:

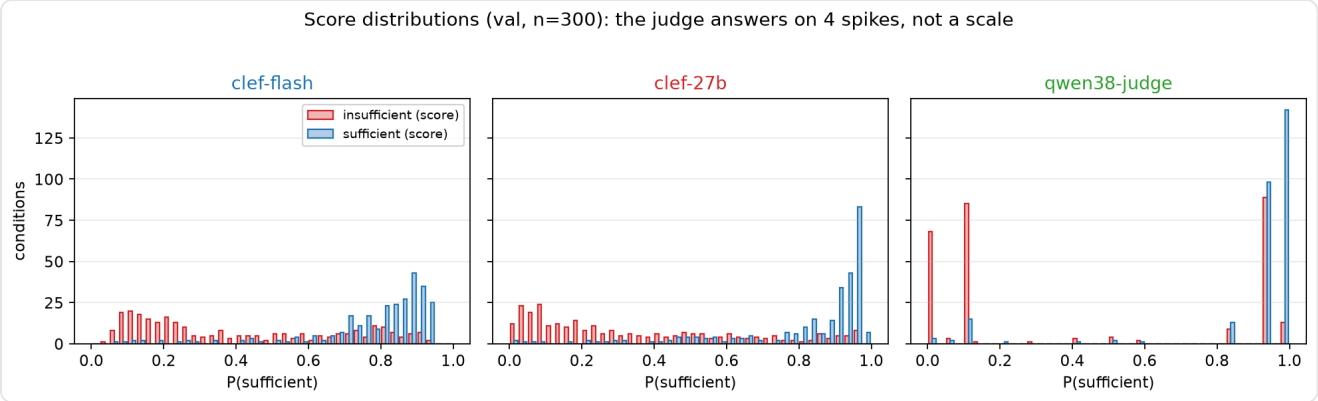
- The **purple "perfect" line** hugs the top-left corner: a flawless judge answers everything worth answering and nothing that isn't.
- The **red dashed diagonal** is the coin flip: to approve twice as many good questions, you're forced to wave through exactly twice as many bad ones. Zero skill — the payoff never outruns the risk.
- **Every real judge sits in between**, and the question is simply *how far above the coin-flip line its curve bows*.

AUROC is just the area of the shaded region under a gate's curve, as a score from 0.50 (coin flip) to 1.00 (perfect). It's the number we quote because it grades a gate's *raw judgement ability* independent of where we happen to set the threshold — the curve lifts up and left as a judge gets better, and AUROC is the single-number summary of that lift.

Two useful gut checks when you see one quoted:

- **0.50 = useless** — pure coin flip. (Our dumb control tests score exactly this, which is how we know the exam itself isn't fooling us.)
- **The gap is what matters, not the decimal.** The difference between 0.83 and 0.87 is *not* "4% better" in plain terms — in live traffic it translates directly into how many more questions the machine can be trusted to handle without a human. Clef-27B's AUROC of 0.87 vs Clef-Flash's 0.83 is a genuinely large, statistically significant capability jump — and on the full exam the statistics are now decisive, not suggestive (§5).

6. The week's key discovery: the chat judge's fake confidence



How to read this chart: each panel shows how one model spread its confidence scores across 556 test cases — blue bars = cases it should approve (evidence sufficient), red = cases it should block. A *good* judge pushes the blue bars to the right and the red bars to the left, with a clean gap between them.

What it says: - **Clef-Flash and Clef-27B (left, middle)** behave like real analysts: they spread their opinions across the whole 0–100% range, and the blue/red separation (their "quality") is what makes a working gate. - **The chat judge (right)** doesn't have opinions — it *shouts*. Almost every answer lands on exactly four values: 0%, 10%, 95%, or 100%. Worse, its 95% answers are right only about half the time — it uses 95% to mean "no idea." That's why its trustworthiness score (ECE) is the worst of every option, and why a safety threshold placed on its scores is a razor's edge. On the full exam the scale of the trap is stark: of 531 cases sent to the judge, **251 came back "95% sure" — and 123 of those were wrong**. - **The fix we shipped:** we built a "translator" that converts the judge's shouty values onto Clef-Flash's honest scale before any threshold sees them (95% → "actually worth ~51%, block"; 100% → "worth ~94%, approve"). Verified live at full scale: **all 251 of the judge's "95%" verdicts — including all 123 confidently-wrong ones — landed at ~51% and were correctly blocked**; zero leaked through untranslated. This rescued the two-step combo from near-uselessness (2% → 19% coverage) and made it the *best-calibrated* option on the board — but still not good enough to beat Clef-27B outright.

7. What the machine costs (and it fits)

	Clef-Flash	Clef-27B
Typical delay per checkpoint	0.17 s	0.54 s
Questions processed per second	~6	~2
Server memory	~36 GB	~53 GB
Cost basis	input only, ~750 tokens/question	same

- Both fit comfortably on the Mac Studio (256 GB) **alongside** the chat models we already run.
- Important operational finding: **running several models at once doesn't add capacity** — throughput stays pinned and requests just queue up. We now schedule the heavy checks serially and design around ~6 questions/second per model, which changes how we size a future rollout.

8. Case files — what a real test case looks like

Numbers aside, here is what one exam question actually looks like: the same question asked about two nearly identical documents — one that contains the answer, one that doesn't. (Every example below is verbatim from this week's runs.)

Case 1 — the easy win (history)

> **Question:** "*Who was in charge of the papal army in the War of Barbastro?*"
>
> **Document A (contains the answer):** "*The legendary religious zeal of the Normans was exercised in religious wars long before the First Crusade… In 1064, during the War of Barbastro,* ***William of Montreuil led the papal army*** *and took a huge booty.*"
>
> **Document B — same document, answer sentence surgically removed:** "*…In 1018, Roger de Tosny travelled to the Iberian Peninsula to carve out a state for himself from Moorish lands, but failed.* ***By then however, there were already Norman mercenaries serving as far away as Trebizond and Georgia.***"

Gate	Doc A (answer present)	Verdict	Doc B (answer removed)	Verdict
Clef-Flash	**93%** sure	✅ answer	**11%** sure	⛔ block
Clef-27B	**97%** sure	✅ answer	**2%** sure	⛔ block
Chat judge	**100%** sure	✅ answer	**0%** sure	⛔ block

All three read the pair like humans would: *the answer is in there* vs *it isn't*. This is the gate working as designed — and the two documents are **the same length and topic** on purpose, so no model can cheat by counting words.

Case 2 — the trap, and the save (health question)

> **Question:** "*Is it possible to become physically addicted to weed?*" **Correct answer:** "*No. Marijuana is not physically addictive, but it is possible to become psychologically addicted to it.*"
>
> **Document A (sufficient):** "***No.*** *Marijuana is not physically addictive, but it is possible to become psychologically addicted to it.*"
>
> **Document B (insufficient — tricky!):** "***You absolutely could.*** *Marijuana is not physically addictive, but it is possible to become psychologically addicted to it.*"

Gate	Doc A	Doc B	What happened
Clef-Flash	90% ✅	73%	correct, twice — and flagged Doc B as *contradictory*
Chat judge	95% ✅	95%	**gave both docs the identical score** — fooled
Two-step combo	✅	⛔	judge's 95% translated to "*really only ~51% trustworthy*" → correctly blocked

This single pair is the project in miniature: one word different ("No" vs "You absolutely could"), the chat judge couldn't tell them apart, and the translation layer we built this week (§6) is what caught the trap. Clef-Flash even *diagnosed* why — it labelled Doc B "contradictory," not just "insufficient."

Case 3 — what "coverage" costs look like in practice

> **Question:** "*…determine if the value of an input is less than what variable?*" (a computer-science question about factorization)

Gate	Confidence	Threshold	Outcome
Clef-Flash	74%	89%	⛔ blocked — "not confident enough"
Clef-27B	91%	93%	⛔ blocked — *so close*
Chat judge	100%	—	saw it clearly, and the combo gate let it answer

This is where auto-answer traffic gets lost today: all three models sort of knew, no single gate crossed its bar. Whether stronger combos can safely harvest these near-misses is exactly what next week's work targets.

9. Recommendation & next week

Ship Clef-27B as the production safety gate. On the full exam it wins on accuracy (AUROC 0.87) and trustworthiness together; its lead over every alternative is now measured on 2,224 test questions and is statistically decisive, and the infrastructure supports it today. Keep Clef-Flash as the budget/lower-latency fallback, and park the two-step combo — correctly engineered now (it is literally the best-calibrated option on the board), but it can't yet justify its complexity. The full exam also delivered a new, clean disqualifier: the chat judge run alone **exceeds our 5% safety budget (5.5%)** — its headline coverage cannot be bought at the risk we accept.

1. **Biggest opportunity:** every option today answers only ~1 in 5 questions it could safely answer. Next we test a purpose-trained "sufficiency probe" that could lift that substantially — coverage, not safety, is where the business value is unlocked.
2. Make the chat judge *earn* its escalations (a smarter "thinking mode" for hard cases only).

3. Build a labelled contradiction test set — the gate's second job (catching documents that disagree with each other) has promising early signs but no proof yet.
4. Escalate the throughput ceiling to the platform team: today the checkpoint caps at ~6 questions/second per model; a second replica or an upstream fix (batching) unblocks rollout scale.

Appendix — full technical detail for anyone who wants it: `FINDINGS.md` (12 sections of measurements and methodology), `REPORT.md` (model-by-model summary), `PLAN-FULL-RUN.md` (how the full run was planned and how the actuals compared), raw run data in `results/full-benchmark-v1/` (13,900 predictions), all charts reproducible via `scripts/make_report_figures.py full-benchmark-v1`, `scripts/make_exec_figures.py full-benchmark-v1` and `scripts/make_roc_explainer.py full-benchmark-v1`.*

Generated 2026-10-14 — full data: results/full-benchmark-v1 (2,780 conditions/arm, zero errors).